

## EXP 7: Temporal Difference Learning using TD(0) Algorithm

### Aim

To implement the TD(0) Temporal Difference Learning algorithm and estimate the value function of states using experience obtained step-by-step.

### Algorithm

1. Initialize the value function for all states.
2. Start from an initial state.
3. Take an action and observe the **reward** and **next state**.
4. Calculate the TD target:
5. Update the current state value:
6. Move to the next state.
7. Repeat until the episode ends.
8. Repeat for multiple episodes until the values converge.

### CODE

```
import gymnasium as gym
import numpy as np
import matplotlib.pyplot as plt
from collections import defaultdict

env = gym.make("Blackjack-v1")
V = defaultdict(float)
alpha = 0.1
gamma = 1.0
episodes = 5000
rewards = []

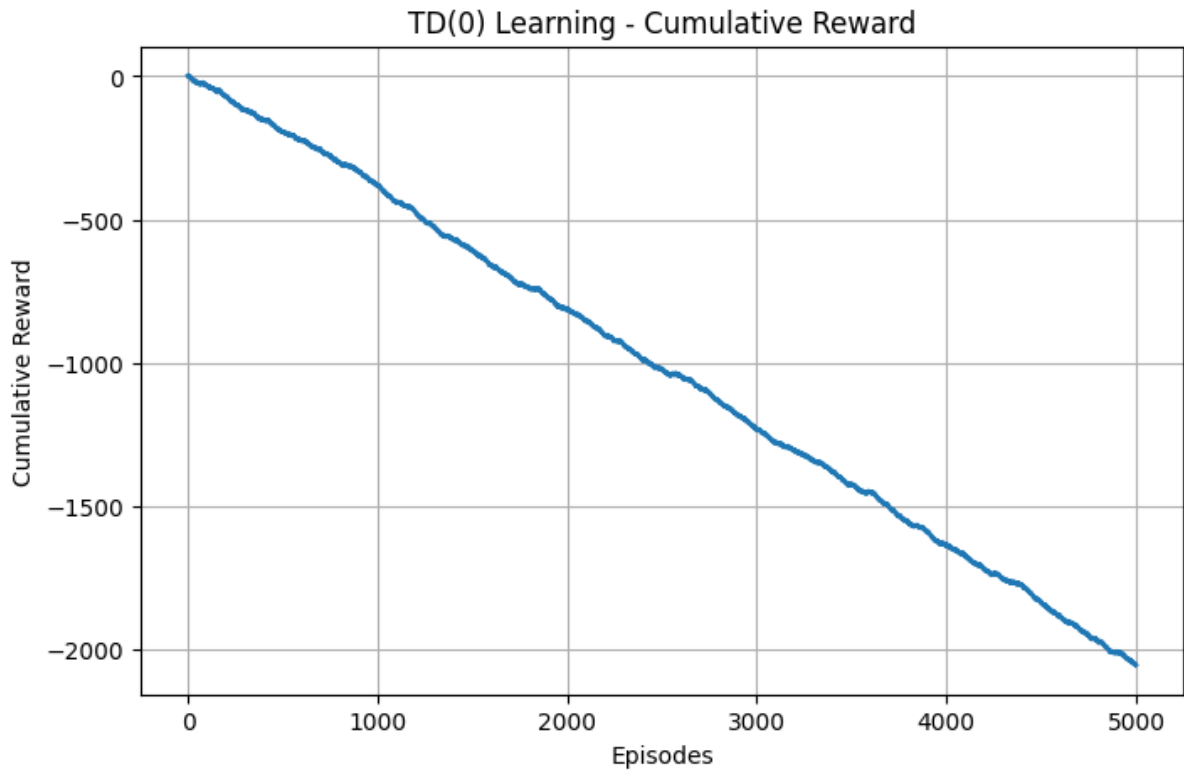
for episode in range(episodes):
    state, _ = env.reset()
    total_reward = 0
    while True:
        action = np.random.choice([0, 1])
```

```

    next_state, reward, done, _, _ = env.step(action)
    if done:
        target = reward
    else:
        target = reward + gamma * V[next_state]
    V[state] += alpha * (target - V[state])
    total_reward += reward
    if done:
        break
    state = next_state
    rewards.append(total_reward)
cumulative_reward = np.cumsum(rewards)
plt.figure(figsize=(8, 5))
plt.plot(
    cumulative_reward,
    linewidth=2
)
plt.xlabel("Episodes")
plt.ylabel("Cumulative Reward")
plt.title("TD(0) Learning - Cumulative Reward")
plt.grid(True)
plt.show()
env.close()
print("TD(0) Learning Completed Successfully.")

```

## OUTPUT



## Result

Thus, the TD(0) algorithm was successfully implemented. The value function was updated after every step using the current observed reward and estimated value of the next state. The algorithm gradually learned the expected value of each state without waiting for the complete episode to finish.